

9. Cobalt forms a range of complex ions. Two of these are $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{CoCl}_4]^{2-}$.

(a) Draw the structures of the $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{CoCl}_4]^{2-}$ ions. [2]

(b) Explain why the $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ ion is coloured. [3]

.....

.....

.....

.....

.....

.....

.....



- (c) A student was given a pink-coloured solution containing $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ ions. Upon addition of hydrochloric acid the solution turned blue as $[\text{CoCl}_4]^{2-}$ ions formed according to the equilibrium shown below.



Aqueous silver nitrate was added to the solution containing $[\text{CoCl}_4]^{2-}$.

State and explain the observation(s) expected.

[4]

.....

.....

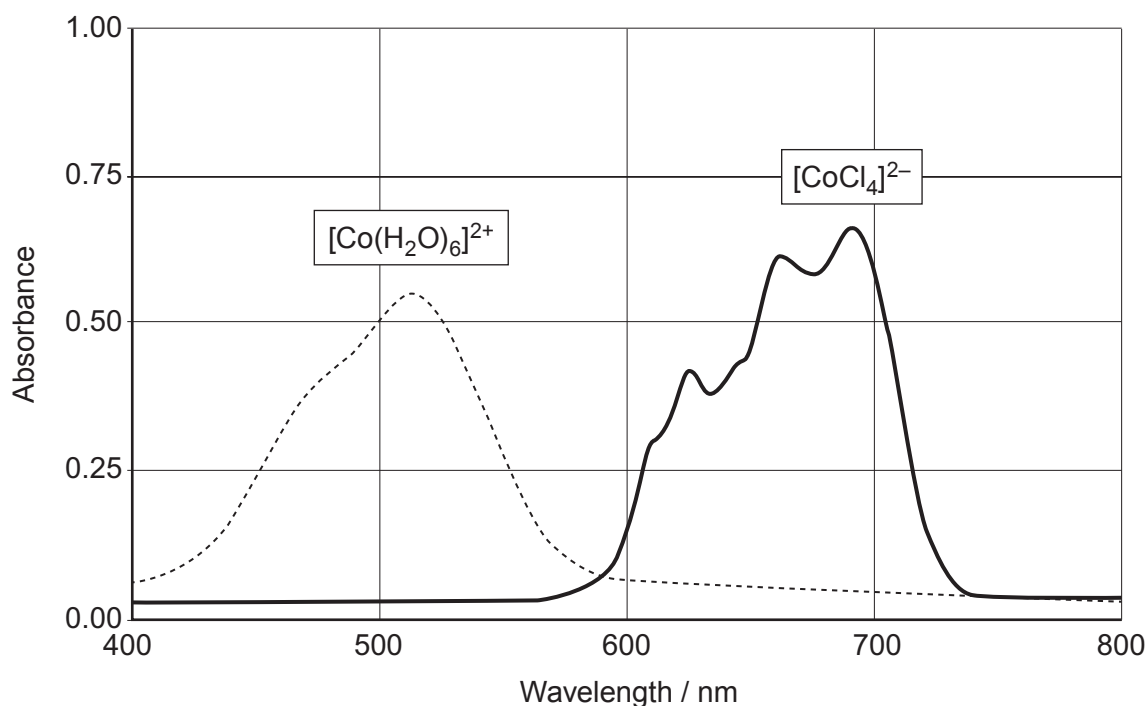
.....

.....

.....



- (d) These two cobalt-containing ions absorb different frequencies of light. The spectra below show the absorbance at different wavelengths of visible light by the two ions.



- (i) The maximum absorbance for the $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ ion occurs at a wavelength of 518 nm. Calculate the energy of this absorbance in kJ mol^{-1} . [3]

Energy = kJ mol^{-1}

- (ii) A colorimeter was used to study the equilibrium between the two ions.

Suggest, giving a reason, an appropriate wavelength to use for the experiment. [1]

.....

.....

.....



- (iii) Write the expression for the equilibrium constant, K_c , for this reaction **giving its unit**. [2]



$K_c =$

Unit

- (iv) A student carries out an experiment starting with separate non-aqueous solutions of $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ and Cl^- .

When the two solutions are mixed the initial concentrations of both $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ and Cl^- in the mixture are $0.720 \text{ mol dm}^{-3}$. When the mixture reaches equilibrium the concentration of the water is $0.744 \text{ mol dm}^{-3}$.

Calculate the value of the equilibrium constant, K_c . [3]

$K_c =$

1410U301
09

18

